

ZHAN'S GOURMET GOODIES

Bakery Process Data Modeling



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I. Normalization

Notations:

- PRIMARY KEY(underlined)
- **FOREIGN KEY**(Bold)

EMPLOYEE(EmployeeID, FirstName, LastName, Street, City, State Zip, Type)

EMPLOYEE_MANAGER(EmployeeID)

EmployeeID is foreign key referencing the EMPLOYEE table (EmployeeID).

EMPLOYEE_OTHER(ManagerID)

ManagerID is foreign key referencing the EMPLOYEE_MANAGER table (EmployeeID).

EMPLOYEE_OTHER_BAKER(EmployeeID, Field)

EmployeeID is foreign key referencing the EMPLOYEE table (EmployeeID).

EMPLOYEE_OTHER_DRIVER(EmployeeID, LicenseNo)

EmployeeID is foreign key referencing the EMPLOYEE table (EmployeeID).

PRODUCT(ProductID, ProductName, Description, Price, Category)

PRODUCT_DRIVER_STOREBAKERY(ProductID, LocationID, EmployeeID)

ProductID is foreign key referencing the PRODUCT table (ProductID).

LocationID is foreign key referencing the LOCATION table (LocationID).

EmployeeID is foreign key referencing the EMPLOYEE table (EmployeeID).

INGREDIENT(IngredientID, IngredientName, ContractPrice, ExpirationDate, **LocationID**)

LocationID is foreign key referencing the LOCATION table (LocationID).

BAKER_INGREDIENT(EmployeeID, IngredientID)

EmployeeID is foreign key referencing the EMPLOYEE table (EmployeeID).

LocationID is foreign key referencing the LOCATION table (LocationID).

SUPPLIER(SupplierID, SupplierName, PhoneNumber, Street, City, State, Zip, EmailAddress)

PROCUREMENT(ProcurementID, **SupplierID**, **IngredientID**, Date, QuantitySupplied, **ManagerID**)

SupplierID is foreign key referencing the SUPPLIER table (SupplierID).

IngredientID is foreign key referencing the INGREDIENT table (IngredientID).

ManagerID is a foreign key referencing the EMPLOYEE_MANAGER table (EmployeeID).

LOCATION(LocationID, LocationName, Street, City, State, Zip)

LOCATION_INDUSTRIALKITCHEN(LocationID, **IngredientID**)

LocationID is foreign key referencing the LOCATION table (LocationID).
IngredientID is foreign key referencing the INGREDIENT table (IngredientID).

LOCATION_STOREBAKERY(LocationID)

LocationID is foreign key referencing the LOCATION table (LocationID).

RECIPE(RecipeID, **ProductID**, **IngredientID**, FromDate)

IngredientID is foreign key referencing INGREDIENT table (IngredientID)

ProductID is foreign key referencing PRODUCT table (ProductID)

II. Create Database:

Drop Database If EXISTS ZhanGourmet;

CREATE DATABASE ZhanGourmet;

USE ZhanGourmet;

#1 Employee Table

```
CREATE TABLE Employee (  
    EmployeeID CHAR(4) NOT NULL,  
    FirstName VARCHAR(20),  
    LastName VARCHAR(20),  
    Street VARCHAR(60),  
    City VARCHAR(30),  
    State CHAR(2),  
    Zip CHAR(5),  
    Type VARCHAR(20),  
    PRIMARY KEY(EmployeeID)  
);
```

#2 Employee_Manager

```
CREATE TABLE Employee_Manager (  
    EmployeeID CHAR(4),  
    PRIMARY KEY (EmployeeID),  
    FOREIGN KEY (EmployeeID) references Employee (EmployeeID)  
);
```

#3 Employee_Other

```
CREATE TABLE Employee_Other (  
    EmployeeID CHAR(4),  
    ManagerID CHAR(4),  
    PRIMARY KEY (EmployeeID),  
    FOREIGN KEY (ManagerID) references Employee_Manager (EmployeeID),
```

```
FOREIGN KEY (EmployeeID) references Employee (EmployeeID)
);
```

```
# 4 Employee_Baker
CREATE TABLE Employee_Baker (
EmployeeID CHAR(4),
PRIMARY KEY (EmployeeID),
FOREIGN KEY (EmployeeID) references Employee_Other (EmployeeID)
);
```

```
# 5 Employee_Driver
CREATE TABLE Employee_Driver (
EmployeeID CHAR(4),
LicenseNo CHAR (15),
PRIMARY KEY (EmployeeID),
FOREIGN KEY (EmployeeID) references employee_Other (EmployeeID)
);
```

```
# 6 Product
CREATE TABLE Product (
ProductID CHAR(6) NOT NULL,
ProductName VARCHAR(30),
Description VARCHAR(255) DEFAULT NULL,
Price FLOAT DEFAULT NULL,
Category VARCHAR(20) DEFAULT NULL,
PRIMARY KEY (ProductID)
);
```

```
# 7 Location
CREATE TABLE Location (
LocationID CHAR(4) NOT NULL,
LocationName VARCHAR(40),
Street VARCHAR(60),
City VARCHAR(30),
State CHAR(2),
Zip CHAR(5),
PRIMARY KEY (LocationID)
);
```

```
# 8 Location_IndustrialKitchen
CREATE TABLE Location_IndustrialKitchen (
LocationID CHAR(4),
PRIMARY KEY (LocationID),
FOREIGN KEY (LocationID) references Location (LocationID)
```

);

9 Location_StoreBakery

```
CREATE TABLE Location_StoreBakery (  
  LocationID CHAR(4),  
  PRIMARY KEY (LocationID),  
  FOREIGN KEY (LocationID) references Location (LocationID)  
);
```

10 Ingredient

```
CREATE TABLE Ingredient (  
  IngredientID CHAR(6) NOT NULL,  
  IngredientName VARCHAR(30),  
  ContractedPrice FLOAT NOT NULL,  
  ExpirationDate DATE NOT NULL,  
  LocationID CHAR(4),  
  PRIMARY KEY (IngredientID),  
  FOREIGN KEY (LocationID) REFERENCES Location_IndustrialKitchen(LocationID)  
);
```

11 Supplier

```
CREATE TABLE Supplier (  
  SupplierID CHAR(4) NOT NULL,  
  SupplierName VARCHAR(40),  
  PhoneNumber VARCHAR(20),  
  Street VARCHAR(60),  
  City VARCHAR(30),  
  State CHAR(2),  
  Zip CHAR(5),  
  EmailAddress VARCHAR(50),  
  PRIMARY KEY (SupplierID)  
);
```

12 Recipe

```
CREATE TABLE Recipe (  
  RecipeID CHAR(4) NOT NULL,  
  ProductID CHAR(6),  
  IngredientID CHAR(4),  
  FromDate DATE,  
  PRIMARY KEY(RecipeID),  
  FOREIGN KEY(ProductID) REFERENCES Product (ProductID),  
  FOREIGN KEY(IngredientID) REFERENCES Ingredient (IngredientID)  
);
```

13 Procurement

```
CREATE TABLE Procurement (  
    ProcurementID CHAR(4),  
    SupplierID CHAR(4) NOT NULL,  
    IngredientID CHAR(6) NOT NULL,  
    Date DATE NOT NULL,  
    QuantitySupplied INT,  
    ManagerID CHAR(4) NOT NULL,  
    PRIMARY KEY (ProcurementID),  
    FOREIGN KEY(SupplierID) REFERENCES Supplier (SupplierID),  
    FOREIGN KEY(IngredientID) REFERENCES Ingredient (IngredientID),  
    FOREIGN KEY(ManagerID) REFERENCES Employee_Manager(EmployeeID)  
);
```

14 Driver_Store_Bakery_Product

```
CREATE TABLE Driver_Store_Bakery_Product (  
    EmployeeID CHAR(4),  
    ProductID CHAR(6),  
    LocationID CHAR(4),  
    PRIMARY KEY (EmployeeID, ProductID, LocationID),  
    FOREIGN KEY (EmployeeID) references Employee_Driver (EmployeeID),  
    FOREIGN KEY (ProductID) references Product (ProductID),  
    FOREIGN KEY (LocationID) references Location_StoreBakery (LocationID)  
);
```

15 Baker_Ingredient

```
CREATE TABLE Baker_Ingredient (  
    EmployeeID CHAR(4),  
    IngredientID CHAR(6),  
    PRIMARY KEY (EmployeeID, IngredientID),  
    FOREIGN KEY (EmployeeID) references Employee_Baker (EmployeeID),  
    FOREIGN KEY (IngredientID) references Ingredient (IngredientID)  
);
```

III. Populate

-- Employee

```
INSERT INTO Employee (EmployeeID, FirstName, LastName, Street, City, State, Zip, Type)  
VALUES  
( '0001', 'John', 'Doe', '123 Main St', 'San Luis Obispo', 'CA', '93401', 'Manager'),
```

```

('0002', 'Jane', 'Doe', '456 Elm St', 'Los Angeles', 'CA', '90001', 'Baker'),
('0003', 'Bob', 'Smith', '789 Ocean Ave', 'Pismo Beach', 'CA', '93449', 'Driver'),
('0004', 'Mike', 'Williams', '123 Downtown St', 'San Luis Obispo', 'CA', '93401', 'Baker'),
('0005', 'Sara', 'Davis', '456 Vine St', 'Paso Robles', 'CA', '93446', 'Driver'),
('0006', 'Tom', 'Anderson', '789 Hollywood Blvd', 'Los Angeles', 'CA', '90001', 'Driver'),
('0007', 'Olivia', 'Garcia', '30 Ocean Blvd', 'Long Beach', 'CA', '90802', 'Baker'),
('0008', 'Daniel', 'Martinez', '123 Main St', 'San Luis Obispo', 'CA', '93401', 'Baker'),
('0009', 'Sophia', 'Lopez', '456 Elm St', 'Los Angeles', 'CA', '90001', 'Manager'),
('0010', 'Liam', 'Taylor', '789 Ocean Ave', 'Pismo Beach', 'CA', '93449', 'Manager'),
('0011', 'Oliver', 'Harris', '123 Downtown St', 'San Luis Obispo', 'CA', '93401', 'Manager'),
('0012', 'Emma', 'Clark', '456 Vine St', 'Paso Robles', 'CA', '93446', 'Manager'),
('0013', 'Maria', 'Garcia', '123 Broadway St', 'Los Angeles', 'CA', '90003', 'Baker'),
('0014', 'Michael', 'Lee', '456 Vine St', 'Los Angeles', 'CA', '90004', 'Baker'),
('0015', 'Jason', 'Frank', '27740 Pacific Coast Hwy', 'Malibu', 'CA', '90265', 'Other'),
('0016', 'Alice', 'Johnson', '123 Apple St', 'San Francisco', 'CA', '94101', 'Other'),
('0017', 'Bob', 'Miller', '456 Orange Ave', 'Los Angeles', 'CA', '90001', 'Other'),
('0020', 'Charlie', 'Brown', '789 Peach Blvd', 'San Diego', 'CA', '92101', 'Other'),
('0021', 'David', 'Davis', '123 Cherry Ln', 'Sacramento', 'CA', '94203', 'Other'),
('0029', 'Eve', 'Martin', '456 Lemon Dr', 'Fresno', 'CA', '93721', 'Other'),
('0030', 'Frank', 'Garcia', '789 Lime Way', 'Long Beach', 'CA', '90802', 'Other'),
('0031', 'Grace', 'Rodriguez', '123 Berry Rd', 'Oakland', 'CA', '94601', 'Other'),
('0032', 'Harry', 'Martinez', '456 Melon St', 'Bakersfield', 'CA', '93301', 'Other'),
('0033', 'Ivy', 'Hernandez', '789 Grape Ave', 'Anaheim', 'CA', '92801', 'Other');

```

-- Employee_Manager

```
INSERT INTO Employee_Manager (EmployeeID)
```

```
VALUES
```

```

('0001'), -- John Doe
('0009'), -- Sophia Lopez
('0010'), -- Liam Taylor
('0011'), -- Oliver Harris
('0012'); -- Emma Clark

```

-- Employee_Other

```
INSERT INTO Employee_Other (EmployeeID, ManagerID)
```

```
VALUES
```

```

('0015', '0001'), -- Jason Frank, managed by John Doe
('0016', '0001'), -- New Employee, managed by John Doe
('0017', '0010'), -- New Employee, managed by Liam Taylor
('0020', '0012'), -- New Employee, managed by Emma Clark
('0021', '0011'), -- New Employee, managed by Oliver Harris
('0029', '0001'), -- New Employee, managed by John Doe
('0030', '0010'), -- New Employee, managed by Liam Taylor
('0031', '0012'), -- New Employee, managed by Emma Clark

```

```
('0032', '0011'), -- New Employee, managed by Oliver Harris
('0033', '0001'); -- New Employee, managed by John Doe
```

```
-- Employee_Baker
INSERT INTO Employee_Baker (EmployeeID)
VALUES
('0015'), -- Jane Doe
('0016'), -- Mike Williams
('0017'), -- Olivia Garcia
('0020'), -- Daniel Martinez
('0029'); -- Maria Garcia
```

```
-- Employee_Driver
INSERT INTO Employee_Driver (EmployeeID, LicenseNo)
VALUES
('0030', 'DL123456'), -- Bob Smith
('0031', 'DL789012'), -- Sara Davis
('0032', 'DL345678'), -- Tom Anderson
('0033', 'DL111111'); -- New Driver
```

```
-- Product
INSERT INTO Product (ProductID, ProductName, Description, Price, Category)
VALUES
('1', 'Croissant', 'Flaky crescent-shaped pastry', 1.99, 'Baked Goods'),
('2', 'Baguette', 'French bread, long thin loaf', 2.49, 'Baked Goods'),
('3', 'Chocolate Chip Cookie', 'Soft cookie with chocolate chips', 0.99, 'Cookies'),
('4', 'Cinnamon Roll', 'Sweet roll with cinnamon', 2.49, 'Baked Goods'),
('5', 'Sugar Cookie', 'Classic sugar cookie', 1.29, 'Cookies'),
('6', 'French Bread', 'Traditional French bread', 2.99, 'Baked Goods'),
('7', 'Brownie', 'Rich chocolate brownie', 1.99, 'Desserts'),
('8', 'Blueberry Muffin', 'Muffin with fresh blueberries', 1.79, 'Baked Goods'),
('9', 'White Bread', 'Classic white bread loaf', 2.29, 'Baked Goods'),
('10', 'Peanut Butter Cookie', 'Cookie with peanut butter flavor', 1.19, 'Cookies');
```

```
-- Location
INSERT INTO Location (LocationID, LocationName, Street, City, State, Zip)
VALUES
('0001', 'KitchenSLO', '123 Main St', 'San Luis Obispo', 'CA', '93401'),
('0002', 'KitchenLA', '456 Elm St', 'Los Angeles', 'CA', '90001'),
('0003', 'Store1', '789 Ocean Ave', 'Pismo Beach', 'CA', '93449'),
('0004', 'Store2', '123 Downtown St', 'San Luis Obispo', 'CA', '93401'),
('0005', 'Store3', '456 Vine St', 'Paso Robles', 'CA', '93446'),
```



```
('0006', 'Store4', '789 Hollywood Blvd', 'Los Angeles', 'CA', '90001'),  
( '0007', 'Store5', '10 Sunset Blvd', 'Los Angeles', 'CA', '90002'),  
( '0008', 'Store6', '20 Wilshire Ave', 'Los Angeles', 'CA', '90003'),  
( '0009', 'Store7', '30 Ocean Blvd', 'Long Beach', 'CA', '90802'),  
( '0010', 'Store8', '40 Lake Street', 'Los Angeles', 'CA', '90804');
```

-- Location_IndustrialKitchen

```
INSERT INTO Location_IndustrialKitchen (LocationID)
```

```
VALUES
```

```
('0001'),  
( '0002'),  
( '0003'),  
( '0004'),  
( '0005');
```

-- Location_StoreBakery

```
INSERT INTO Location_StoreBakery (LocationID)
```

```
VALUES
```

```
('0007'),  
( '0008'),  
( '0009'),  
( '0010');
```

-- Ingredient

```
INSERT INTO Ingredient (IngredientID, IngredientName, ContractedPrice, ExpirationDate)
```

```
VALUES
```

```
('1', 'Flour', 0.15, '2023-12-01'),  
( '2', 'Butter', 0.99, '2023-11-15'),  
( '3', 'Chocolate Chips', 1.49, '2024-01-15'),  
( '4', 'Sugar', 0.20, '2024-02-01'),  
( '5', 'Eggs', 0.30, '2023-12-10'),  
( '6', 'Vanilla Extract', 0.25, '2024-01-20'),  
( '7', 'Milk', 0.35, '2023-12-15'),  
( '8', 'Yeast', 0.10, '2024-03-01'),  
( '9', 'Cocoa Powder', 0.40, '2024-03-15'),  
( '10', 'Cream Cheese', 0.90, '2024-06-01'),  
( '11', 'Blueberries', 0.95, '2024-07-15'),  
( '12', 'Oats', 0.25, '2024-10-15'),  
( '13', 'Cinnamon', 0.15, '2024-11-15'),  
( '14', 'Powdered Sugar', 0.50, '2024-12-01'),  
( '15', 'Peanut Butter', 1.20, '2024-12-15'),  
( '16', 'Buttermilk', 0.45, '2025-01-01'),  
( '17', 'Peanuts', 0.75, '2025-03-01'),  
( '18', 'Baking Soda', 0.20, '2027-03-01'),
```

```
('19', 'Palm Oil', 0.75, '2025-03-01'),  
( '20', 'Baking Soda', 0.20, '2027-03-01');
```

-- Supplier

```
INSERT INTO Supplier (SupplierID, SupplierName, PhoneNumber, Street, City, State, Zip,  
EmailAddress)  
VALUES  
( '0001', 'Flour Co', '2125551234', '456 Green Rd', 'Los Angeles', 'CA', '90001', 'info@flourco.com'),  
( '0002', 'Butter Inc', '6465555678', '789 Hollow Ln', 'Los Angeles', 'CA', '90001',  
'contact@butterinc.com'),  
( '0003', 'Chocolate Heaven', '8001234567', '123 Cane St', 'Bakersfield', 'CA', '93301',  
'info@chocolateheaven.com'),  
( '0004', 'Dairy Delights', '8885551234', '456 Greenwich Way', 'Bakersfield', 'CA', '93302',  
'contact@dairydelights.com');
```

-- Recipe

```
INSERT INTO Recipe (RecipeID, ProductID, IngredientID, FromDate)  
VALUES  
( '2001', '1', '1', '2023-11-16'), -- Croissant: Flour  
( '2002', '1', '2', '2023-11-17'), -- Croissant: Butter  
( '2003', '1', '4', '2023-11-18'), -- Croissant: Sugar  
( '2004', '2', '1', '2023-11-19'), -- Baguette: Flour  
( '2005', '2', '8', '2023-11-20'), -- Baguette: Yeast  
( '2006', '2', '4', '2023-11-21'), -- Baguette: Sugar  
( '2007', '3', '3', '2023-11-22'), -- Chocolate Chip Cookie: Chocolate Chips  
( '2008', '3', '4', '2023-11-23'), -- Chocolate Chip Cookie: Sugar  
( '2009', '3', '1', '2023-11-24'), -- Chocolate Chip Cookie: Flour  
( '2010', '4', '13', '2023-11-25'), -- Cinnamon Roll: Cinnamon  
( '2011', '4', '4', '2023-11-26'), -- Cinnamon Roll: Sugar  
( '2012', '4', '1', '2023-11-27'), -- Cinnamon Roll: Flour  
( '2013', '5', '4', '2023-11-28'), -- Sugar Cookie: Sugar  
( '2014', '5', '1', '2023-11-29'), -- Sugar Cookie: Flour  
( '2015', '5', '5', '2023-11-30'), -- Sugar Cookie: Eggs  
( '2018', '6', '1', '2023-12-01'), -- French Bread: Flour  
( '2019', '6', '8', '2023-12-02'), -- French Bread: Yeast  
( '2020', '6', '4', '2023-12-03'), -- French Bread: Sugar  
( '2021', '7', '9', '2023-12-04'), -- Brownie: Cocoa Powder  
( '2022', '7', '4', '2023-12-05'), -- Brownie: Sugar  
( '2023', '7', '1', '2023-12-06'), -- Brownie: Flour  
( '2024', '8', '11', '2023-12-07'), -- Blueberry Muffin: Blueberries  
( '2025', '8', '1', '2023-12-08'), -- Blueberry Muffin: Flour  
( '2026', '8', '4', '2023-12-09'), -- Blueberry Muffin: Sugar  
( '2027', '9', '1', '2023-12-10'), -- White Bread: Flour
```

```
('2028', '9', '4', '2023-12-11'), -- White Bread: Sugar
('2029', '9', '8', '2023-12-12'), -- White Bread: Yeast
('2030', '10', '15', '2023-12-13'), -- Peanut Butter Cookie: Peanut Butter
('2031', '10', '4', '2023-12-14'), -- Peanut Butter Cookie: Sugar
('2032', '10', '1', '2023-12-15'); -- Peanut Butter Cookie: Flour
```

-- Procurement

```
INSERT INTO Procurement (ProcurementID, SupplierID, IngredientID, Date, QuantitySupplied,
ManagerID)
```

VALUES

```
('1001', '0001', '5', '2023-11-16', 100, '0001'), -- John Doe
('1002', '0002', '12', '2023-11-17', 150, '0009'), -- Sophia Lopez
('1003', '0003', '8', '2023-11-18', 200, '0010'), -- Liam Taylor
('1004', '0004', '16', '2023-11-19', 120, '0011'), -- Oliver Harris
('1005', '0001', '3', '2023-11-20', 180, '0012'), -- Emma Clark
('1006', '0002', '10', '2023-11-21', 90, '0001'), -- John Doe
('1007', '0003', '7', '2023-11-22', 220, '0009'), -- Sophia Lopez
('1008', '0004', '19', '2023-11-23', 130, '0010'), -- Liam Taylor
('1009', '0001', '14', '2023-11-24', 170, '0011'), -- Oliver Harris
('1010', '0002', '6', '2023-11-25', 110, '0012'); -- Emma Clark
```

-- Driver_Store_Bakery_Product

```
INSERT INTO Driver_Store_Bakery_Product (EmployeeID, ProductID, LocationID)
```

VALUES

```
('0030', '1', '0007'), -- Driver: Bob Smith, Product: Croissant, Location: 0007
('0031', '1', '0008'), -- Driver: Sara Davis, Product: Croissant, Location: 0008
('0032', '1', '0009'), -- Driver: Tom Anderson, Product: Croissant, Location: 0009
('0030', '2', '0007'), -- Driver: Bob Smith, Product: Baguette, Location: 0007
('0031', '2', '0008'), -- Driver: Sara Davis, Product: Baguette, Location: 0008
('0032', '2', '0009'), -- Driver: Tom Anderson, Product: Baguette, Location: 0009
('0030', '3', '0007'), -- Driver: Bob Smith, Product: Chocolate Chip Cookie, Location: 0007
('0031', '3', '0008'), -- Driver: Sara Davis, Product: Chocolate Chip Cookie, Location: 0008
('0032', '3', '0009'), -- Driver: Tom Anderson, Product: Chocolate Chip Cookie, Location: 0009
('0030', '4', '0007'), -- Driver: Bob Smith, Product: Cinnamon Roll, Location: 0007
('0031', '4', '0008'), -- Driver: Sara Davis, Product: Cinnamon Roll, Location: 0008
('0032', '4', '0009'), -- Driver: Tom Anderson, Product: Cinnamon Roll, Location: 0009
('0031', '5', '0008'), -- Driver: Sara Davis, Product: Sugar Cookie, Location: 0008
('0032', '5', '0009'), -- Driver: Tom Anderson, Product: Sugar Cookie, Location: 0009
('0030', '6', '0007'), -- Driver: Bob Smith, Product: French Bread, Location: 0007
('0031', '6', '0008'), -- Driver: Sara Davis, Product: French Bread, Location: 0008
('0032', '6', '0009'), -- Driver: Tom Anderson, Product: French Bread, Location: 0009
('0030', '7', '0007'), -- Driver: Bob Smith, Product: Brownie, Location: 0007
('0031', '7', '0008'), -- Driver: Sara Davis, Product: Brownie, Location: 0008
```

```

('0032', '7', '0009'), -- Driver: Tom Anderson, Product: Brownie, Location: 0009
('0031', '8', '0008'), -- Driver: Sara Davis, Product: Blueberry Muffin, Location: 0008
('0032', '8', '0009'), -- Driver: Tom Anderson, Product: Blueberry Muffin, Location: 0009
('0030', '9', '0007'), -- Driver: Bob Smith, Product: White Bread, Location: 0007
('0031', '9', '0008'), -- Driver: Sara Davis, Product: White Bread, Location: 0008
('0032', '9', '0009'), -- Driver: Tom Anderson, Product: White Bread, Location: 0009
('0031', '10', '0008'), -- Driver: Sara Davis, Product: Peanut Butter Cookie, Location: 0008
('0032', '10', '0009'), -- Driver: Tom Anderson, Product: Peanut Butter Cookie, Location: 0009

```

-- Baker_Ingredient

```

INSERT INTO Baker_Ingredient (EmployeeID, IngredientID)

```

VALUES

```

('0015', '1'), -- Baker Jane Doe, Flour
('0015', '2'), -- Baker Jane Doe, Butter
('0015', '4'), -- Baker Jane Doe, Sugar
('0016', '4'), -- Baker Mike Williams, Sugar
('0016', '6'), -- Baker Mike Williams, Vanilla Extract
('0016', '7'), -- Baker Mike Williams, Milk
('0017', '3'), -- Baker Olivia Garcia, Chocolate Chips
('0017', '9'), -- Baker Olivia Garcia, Cocoa Powder
('0017', '13'), -- Baker Olivia Garcia, Cinnamon
('0020', '4'), -- Baker Daniel Martinez, Sugar
('0020', '6'), -- Baker Daniel Martinez, Vanilla Extract
('0020', '7'), -- Baker Daniel Martinez, Milk
('0029', '5'), -- Baker Maria Garcia, Eggs
('0029', '10'), -- Baker Maria Garcia, Cream Cheese
('0029', '12'), -- Baker Maria Garcia, Oats
('0029', '14'), -- Baker Maria Garcia, Powdered Sugar
('0029', '15'), -- Baker Maria Garcia, Peanut Butter

```

IV. Testing

Create at least 5 queries that generate meaningful reports for a business user of this system for the intended process. Queries need to span multiple tables. At least 3 of them need to include grouping functions. At least 2 of them need to have subqueries.

1. (group by)Total Procurement by Supplier: This query will calculate the total quantity of ingredients supplied by each supplier.

```
SELECT Supplier.SupplierName, SUM(Procurement.QuantitySupplied) AS TotalQuantitySupplied
FROM Supplier
JOIN Procurement ON Supplier.SupplierID = Procurement.SupplierID
GROUP BY Supplier.SupplierName;
```

	SupplierName	TotalQuantitySupplied
▶	Flour Co	450
	Butter Inc	350
	Chocolate Heaven	420
	Dairy Delights	250

- (group by) Identifying Premium Products in High-Demand Locations: This query finds premium products in Location_StoreBakery outlets by comparing each product's price to the average price of all products across these locations. Products priced above this average are considered premium and potentially in high demand.

```
SELECT P.ProductID, P.ProductName, P.Price, P.Category
FROM Product P JOIN Driver_Store_Bakery_Product DSBP ON P.ProductID = DSBP.ProductID
JOIN Location_StoreBakery LSB ON DSBP.LocationID = LSB.LocationID
GROUP BY P.ProductID, P.ProductName, P.Price, P.Category;
```

ProductID	ProductName	Price	Category
1	Croissant	1.99	Baked Goods
2	Baguette	2.49	Baked Goods
3	Chocolate Chip Cookie	0.99	Cookies
4	Cinnamon Roll	2.49	Baked Goods
6	French Bread	2.99	Baked Goods
7	Pretzel	1.99	Snacks

- (subquery) Top Suppliers Based on Quantity Supplied: This query identifies the top suppliers based on the quantity of ingredients they have supplied. It uses a subquery to find the average quantity supplied across all suppliers, and then lists suppliers who have supplied more than this average.

```
SELECT Supplier.SupplierName, SUM(Procurement.QuantitySupplied) AS TotalQuantity
FROM Supplier
JOIN Procurement ON Supplier.SupplierID = Procurement.SupplierID
GROUP BY Supplier.SupplierName
HAVING SUM(Procurement.QuantitySupplied) > (
```

```

SELECT AVG(TotalQuantity)
FROM (
    SELECT SUM(QuantitySupplied) AS TotalQuantity
    FROM Procurement
    GROUP BY SupplierID
) AS SubQuery
);

```

	SupplierName	TotalQuantity
▶	Flour Co	450
	Chocolate Heaven	420

4. (group by) Average Ingredient Contracted Price by Supplier: This query will calculate the average contracted price of ingredients supplied by each supplier. This information can help in assessing the pricing policies of different suppliers and managing costs effectively.

```

SELECT Supplier.SupplierName, AVG(Ingredient.ContractPrice) AS AvgContractedPrice
FROM Ingredient JOIN Procurement ON Ingredient.IngredientID = Procurement.IngredientID
JOIN Supplier ON Procurement.SupplierID = Supplier.SupplierID
GROUP BY Supplier.SupplierName;

```

SupplierName	AvgContractedPrice
Flour Co	0.7633333404858907
Butter Inc	0.4666666587193807
Chocolate Heaven	0.22499999776482582
Dairy Delights	0.5999999940395355

5. (subquery)Average Product Price by Category with Count of Products Exceeding the Average Price: This query calculates the average price of products in each category and counts how many products in each category have a price above the category's average.

```

SELECT
    p1.Category,
    AVG(p1.Price) AS AvgPrice,
    (
        SELECT COUNT(*)
        FROM Product p2
        WHERE p2.Category = p1.Category AND p2.Price > AVG(p1.Price)
    ) AS CountAboveAvgPrice
FROM
    Product p1

```

GROUP BY
p1.Category;

	Category	AvgPrice	CountAboveAvgPrice
▶	Baked Goods	2.3399999936421714	3
	Cookies	1.15666667620341	2